

**LOADCHARTS
TMS875B**

85% STABILITY

74180
SERIAL NUMBER

NOTES FOR LIFTING CAPACITIES

GENERAL:

1. Rated loads as shown on lift chart pertain to this machine as originally manufactured and equipped. Modifications to the machine or use of optional equipment other than that specified can result in a reduction of capacity.
2. Construction equipment can be hazardous if improperly operated or maintained. Operation and maintenance of this machine shall be in compliance with the information in the Operator's and Safety Handbook, Service Manual and Parts Manual supplied with this machine. If these manuals are missing, order replacements from the manufacturer through the distributor.
3. The operator and other personnel associated with machine shall fully acquaint themselves with the latest American National Safety Standards (ASME/ANSI) for cranes.

SETUP:

1. The machine shall be level and on a firm supporting surface. Depending on the nature of the supporting surface, it may be necessary to have structural supports under the outrigger floats or tires to spread the load to a larger bearing surface.
2. For outrigger operation, all outriggers shall be properly extended with tires raised free of crane weight before operating the boom or lifting loads.
3. When machine is equipped with center front stabilizer, the front stabilizer shall be set in accordance with instructions in Operator's & Safety Handbook.
4. When equipped with removable and/or extendible counterweight, the proper counterweight shall be installed and fully extended before and during operation.
5. If crane is approved by the manufacturer for on-rubber lifting, tires shall be inflated to the recommended pressure before lifting on rubber.
6. With certain boom and hoist tackle combinations, maximum capacities may not be obtainable with standard cable lengths.
7. Unless approved by the crane manufacturer, do not travel with boom extension or jib erected. Refer to the Operator's and Safety Handbook for job-site travel information.

OPERATION:

1. Rated loads at rated radius shall not be exceeded. Do not attempt to tip the machine to determine allowable loads. For clamshell or concrete bucket operation, weight of bucket and load must not exceed 80% of rated lifting capacities.
2. All rated loads have been tested to and meet the requirements of SAE J1063 - Cantilevered Boom Crane Structures - Method of Test, and do not exceed 85% of the tipping load on outriggers as determined by SAE J765 Crane Stability Test Code.
3. Rated loads include the weight of hookblock, slings and auxiliary lifting devices and their weights shall be subtracted from the listed rating to obtain the net load to be lifted. When more than the minimum required hoist reeving is used, the additional rope weight shall be considered part of the load to be lifted.
4. Load ratings are based on freely suspended loads. No attempt shall be made to move a load horizontally on the ground in any direction.
5. The maximum in-service wind speed is 20 m.p.h. It is recommended when wind velocity is above 20 m.p.h., rated loads and boom lengths shall be appropriately reduced. For machines not in-service, the main boom should be retracted and lowered with the swing brake set in wind velocities over 30 m.p.h.
6. Rated loads are for lift crane service only.
7. Do not operate at a radius or boom length where capacities are not listed. At these positions, the machine may overturn without any load on the hook.
8. The maximum load which can be telescoped is not definable because of variations in loadings and crane maintenance, but it is safe to attempt retraction and extension of the boom within the limits of the capacity chart.
9. When the boom length or lift radius or both are between values listed, the smallest load shown at either the next larger radius or next longer or shorter boom length shall be used.
10. For safe operation, the user shall make due allowances for his particular job conditions, such as: soft or uneven ground, out of level conditions, high winds, side loads, pendulum action, jerking or sudden stopping of loads, hazardous conditions, experience of personnel, two machine lifts, traveling with loads, electric wires, etc. Side pull on boom or jib is extremely dangerous.
11. If machine is equipped with individually controlled powered boom sections, the boom sections must be extended equally at all times.
12. Never handle personnel with this machine unless the requirements of the applicable national, state, and local regulations and safety codes are met.
13. Keep load handling devices a minimum of 42 inches below boom head at all times.
14. The boom angle before loading should be greater than the loaded boom angle to account for deflection.
15. Capacities appearing above the bold line are based on structural strength and tipping should not be relied upon as a capacity limitation.
16. Capacities for the 40 ft. boom length shall be lifted with boom fully retracted. If boom is not fully retracted, capacities shall not exceed those shown for the 45 ft. boom length.
17. Radii less than 30 ft. not recommended when lifting over front of machine.
18. Do not lift loads when boom is fully lowered. The Load Moment Indicator (LMI) senses pressure and will not provide warnings or lockout. The crane can become overloaded if lift cylinder(s) is fully retracted.
19. The maximum outrigger pad load is 101,336 lb.

DEFINITIONS:

1. Operating Radius: Horizontal distance from a projection of the axis of rotation to the supporting surface before loading to the center of the vertical hoist line or tackle with load applied.
2. Loaded Boom Angle (Shown in Parenthesis on Main Boom Capacity Chart): is the angle between the boom base section and the horizontal, after lifting the rated load at the rated radius with the rated boom length.
3. Working Area: Areas measured in a circular arc about the center line of rotation as shown on the working area diagram.
4. Freely Suspended Load: Load hanging free with no direct external force applied except by the lift cable.
5. Side Load: Horizontal force applied to the lifted load either on the ground or in the air.

**RATED LIFTING CAPACITIES IN POUNDS WITH COUNTERWEIGHT
40 FT. - 125 FT. BOOM**

ON OUTRIGGERS FULLY EXTENDED - 360°

Radius in Feet	#0001									
	Main Boom Length in Feet									
	40	45	55	65	75	85	95	105	115	125
10	150,000 (72)	115,000 (74.5)	113,500 (77.5)	84,650 (80)						
12	129,000 (68.5)	115,000 (71.5)	113,500 (75.5)	84,650 (78)	55,450 (80)					
15	101,000 (64)	100,500 (67.5)	99,750 (72)	81,200 (75.5)	55,450 (77.5)	55,450 (79.5)	49,900 (80.5)			
20	72,250 (55)	71,650 (60)	70,850 (66.5)	62,260 (70.5)	55,450 (73.5)	55,450 (76)	49,900 (78)	38,800 (79.5)	27,500 (80.5)	23,000 (81)
25	54,600 (44.5)	54,000 (52)	53,250 (60.5)	49,850 (65.5)	46,600 (69.5)	44,800 (72.5)	42,000 (75)	38,800 (77)	27,500 (78.5)	23,000 (80)
30	43,000 (31)	42,450 (42.5)	41,700 (54)	41,150 (60.5)	38,500 (65.5)	37,000 (69)	35,850 (71.5)	33,750 (74)	25,700 (76)	23,000 (77.5)
35		34,250 (30)	33,550 (46.5)	33,000 (55)	32,450 (61)	31,100 (65)	30,150 (68.5)	28,450 (71)	21,900 (73)	23,000 (75)
40	See Note 16		27,100 (38)	26,150 (49)	26,650 (56)	26,600 (61)	25,800 (65)	24,350 (68)	18,900 (70.5)	21,050 (72.5)
45			21,200 (27)	20,400 (42.5)	20,900 (51)	21,900 (57)	22,300 (61.5)	21,100 (65)	16,500 (68)	18,500 (70)
50				16,150 (35)	16,650 (46)	17,600 (52.5)	18,550 (58)	18,500 (62)	14,650 (65)	16,350 (67.5)
60					10,800 (32.5)	11,650 (43)	12,550 (50)	13,050 (55)	11,750 (59)	13,000 (62)
70						7,880 (30.5)	8,630 (41)	9,110 (47.5)	9,590 (52.5)	10,050 (56.5)
80							5,860 (29)	6,310 (39)	6,770 (45.5)	7,210 (50.5)
90								4,240 (28)	4,680 (37.5)	5,090 (44)
100									3,090 (27)	3,460 (36)
110										2,160 (25.5)
Minimum boom angle (degrees) for indicated length (no load)										0
Maximum boom length (ft.) at 0 degree boom angle (no load)										125

Note: () Boom angles are in degrees.

#LMI operating code. Refer to LMI manual for instructions.

Lifting Capacities On Outriggers - 360 Degrees At Zero Degree Boom Angle										
Boom Angle	Main Boom Length in Feet									
	40	45	55	65	75	85	95	105	115	125
0°	30,400 (33.7)	24,900 (39.1)	17,600 (49.1)	10,750 (59.1)	7,420 (69.1)	5,430 (79.1)	3,960 (89.1)	2,830 (99.1)	1,930 (109.1)	1,230 (118.7)

Note: () Reference radii in feet.

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32 FT. - 56 FT. TELE EXTENSION ON OUTRIGGERS FULLY EXTENDED - 360° WITH COUNTERWEIGHT

Radius in Feet	32 ft. LENGTH			44 ft. LENGTH			56 ft. LENGTH		
	#0021	#0022	#0023	#0031	#0032	#0033	#0041	#0042	#0043
	2° OFFSET	15° OFFSET	30° OFFSET	2° OFFSET	15° OFFSET	30° OFFSET	2° OFFSET	15° OFFSET	30° OFFSET
25	*16,000 (81.6)								
30	16,000 (81)			*10,000 (81.6)					
35	16,000 (79)	*12,300 (81.6)		10,000 (81)			*7,500 (81.6)		
40	15,800 (77)	11,600 (80.5)	*9,000 (81.6)	10,000 (79)	9,000 (81.5)		7,500 (81)		
45	15,600 (75.5)	11,000 (78.5)	8,700 (80)	10,000 (77.5)	8,750 (79.5)		7,500 (79.5)		
50	14,650 (73.5)	10,500 (76.5)	8,500 (78)	9,000 (75.5)	8,300 (78)	6,000 (81.5)	7,500 (78)	6,000 (80)	
60	12,850 (69.5)	9,500 (72.5)	8,000 (74)	8,400 (72)	7,500 (74.5)	5,700 (77.5)	7,500 (74.5)	5,900 (76.5)	4,700 (80.5)
70	9,880 (65.5)	8,600 (68.5)	7,300 (70)	7,800 (68)	6,850 (70.5)	5,400 (74)	6,600 (71)	5,550 (73.5)	4,450 (77)
80	6,900 (61)	6,900 (64)	6,600 (65.5)	6,300 (64.5)	6,300 (66.5)	5,000 (70)	6,100 (67.5)	5,100 (70)	4,150 (73.5)
90	4,690 (56.5)	4,690 (59.5)	4,690 (61)	5,190 (60.5)	5,190 (62.5)	4,800 (65.5)	5,500 (64)	4,700 (66)	3,900 (69.5)
100	2,980 (52)	2,980 (55)	2,980 (56)	4,430 (56)	4,430 (58.5)	4,430 (61.5)	4,500 (60)	4,350 (62.5)	3,700 (66)
110	1,610 (46.5)	1,610 (49.5)	1,610 (50.5)	2,890 (52)	2,890 (54)	2,890 (56.5)	3,730 (56)	3,730 (58.5)	3,500 (61.5)
120				1,620 (47)	1,620 (49)	1,620 (51.5)	2,510 (52)	2,510 (54)	2,510 (57.5)
130							1,460 (47.5)	1,460 (49.5)	1,460 (52.5)

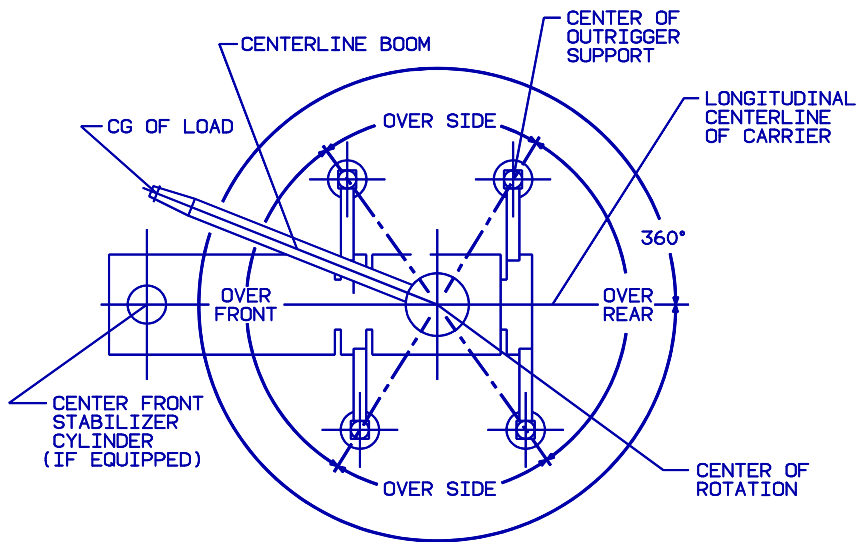
NOTE: () Boom angles are in degrees.

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*This capacity is based upon maximum boom angle.

#LMI operating code. Refer to LMI manual for instructions.

1. All capacities above the bold line are based on structural strength of boom extension.
2. 32 ft., 44 ft. and 56 ft. boom extension lengths may be used for double or single line lifting service.
3. Radii listed are for a fully extended boom with the boom extension erected. For main boom lengths less than fully extended, the rated loads are determined by boom angle. Use only the column which corresponds to the boom extension length and offset for which the machine is configured. For boom angles not shown, use the rating of the next lower boom angle.
WARNING: Operation of this machine with heavier loads than the capacities listed is strictly prohibited. Machine tipping with boom extension occurs rapidly and without advance warning.
4. Boom angle is the angle above or below horizontal of the longitudinal axis of the boom base section after lifting rated load.
5. Capacities listed are with outriggers fully extended and vertical jacks set only.
6. 32 ft. TELE OFFSETTABLE BOOM EXTENSION WARNING: For main boom length greater than 85 ft. with 32 ft. tele. boom extension in working position, the boom angle must not be less than 40° since loss of stability will occur causing a tipping condition. The boom angle is not restricted for main boom length equal to or less than 85 ft. This warning also applies for boom extension erection purposes.
44 ft. TELE OFFSETTABLE BOOM EXTENSION WARNING: For main boom length greater than 85 ft. with 44 ft. tele. boom extension in working position, the boom angle must not be less than 41° since loss of stability will occur causing a tipping condition. The boom angle is not restricted for main boom length equal to or less than 85 ft. This warning also applies for boom extension erection purposes.
56 ft. TELE OFFSETTABLE BOOM EXTENSION WARNING: For main boom length greater than 85 ft. with 56 ft. tele. boom extension in working position, the boom angle must not be less than 42° since loss of stability will occur causing a tipping condition. The boom angle is not restricted for main boom length equal to or less than 85 ft. This warning also applies for boom extension erection purposes.



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BOLD LINES DETERMINE THE LIMITING POSITION OF ANY
LOAD FOR OPERATION WITHIN WORKING AREAS INDICATED
WORKING AREA DIAGRAM

LINE PULLS AND REEVING INFORMATION

HOISTS	CABLE SPECS.	PERMISSIBLE LINE PULLS
Main & Aux. Model 30	3/4" (19 mm) 18x19 Class or 35x7 Rotation Resistant Min. Breaking Strength 64,600 lb	12,920 lb

The approximate weight of 3/4" rope is 1.5 lb/ft.

WEIGHT REDUCTIONS FOR LOAD HANDLING DEVICES

32 FT. BOOM EXTENSION	
*Stowed -	821 lb
*Erected -	6,794 lb
32 FT. - 56 FT. TELE. BOOM EXTENSION	
*Stowed -	1,062 lb
*Erected (retracted) -	8,514 lb
*Erected (extended) -	11,136 lb

*Reduction of main boom capacities

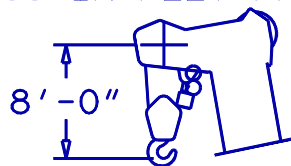
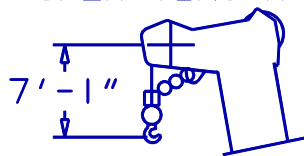
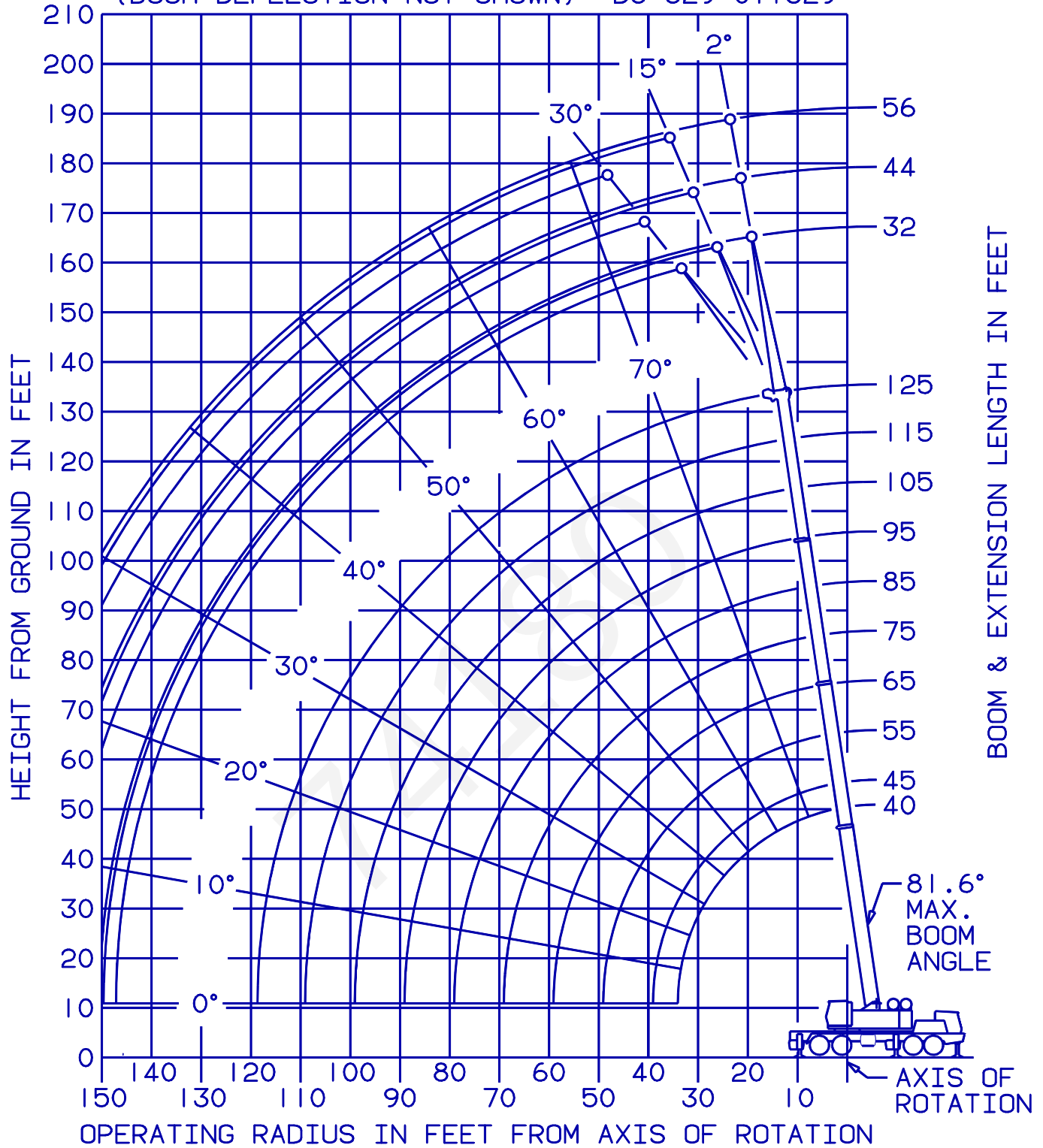
AUXILIARY BOOM HEAD	178 lb
HOOKBLOCKS and HEADACHE BALLS:	
80 ton, 6 sheave	1,930 lb+
45 ton, 3 sheave	1,095 lb+
15 ton, 1 sheave	380 lb+
10 ton headache ball	560 lb+
7.5 ton headache ball	338 lb+

+Refer to rating plate for actual weight.

When lifting over swingaway and/or jib combinations, deduct total weight of all load handling devices reeved over main boom nose directly from swingaway or jib capacity.

NOTE: All load handling devices and boom attachments are considered part of the load and suitable allowances MUST BE MADE for their combined weights. Weights are for Grove furnished equipment.

WORKING RANGE DIAGRAM (BOOM DEFLECTION NOT SHOWN) D6-829-011329



DIMENSIONS ARE FOR LARGEST GROVE FURNISHED HOOK BLOCK AND HEADACHE BALL, WITH ANTI-TWO BLOCK ACTIVATED.